NMIMS MPSTME · B.TECH COMPUTER ENGINEERING

Digital Image Watermarking

A hybrid DWT–DCT–SVD approach — extended evaluation on 100 images, controls, technology comparison and a video feasibility test

CLASSICAL SIGNAL PROCESSING — NO MACHINE LEARNING / DEEP LEARNING

Krishang Vikram Singh
Roll No. B106 · Batch C1
Subject: Image and Video Processing (IVP) · First submission July 2026 · extended edition October 2026

Copying an image takes a second — proving who owns it is the hard part

Imperceptible

The watermark must not visibly change the image. Measured with PSNR and SSIM against the **unwatermarked original**.

Robust

It must survive compression, noise, filtering and geometric change. Measured with NC / ZNCC / BER of the recovered logo.

Explainable

DWT, DCT and SVD are closed-form and verifiable by hand — no black-box model. A deliberate scope choice, not a claim of superiority.

WHAT THIS EXTENDED WORK ADDS

- **100 unique images** instead of two, with seeded attacks, status logging and 43 unit tests
- **Controls** that ask whether a high NC really proves the watermark is present
- **Technology comparison** (libraries and an existing watermarking package) and an exploratory **video** test

Six objectives — three inherited from the first submission, three new

#	Objective	Status
1	Implement the hybrid DWT–DCT–SVD embed/extract pipeline	inherited
2	Quantify imperceptibility (PSNR, SSIM)	inherited · now 100 images
3	Quantify robustness (NC, BER + ZNCC) under 14 attacks and a no-attack reference	inherited · now 100 images
4	Characterise the α trade-off	re-done, 4 attacks
5	Find failure modes — including false positives (controls)	**new**
6	Compare with libraries and an existing package; test video	**new**

Scope boundaries

- 8-bit **grey-scale** images (colour only via luminance)
- **No** machine / deep learning
- Scheme is **non-blind**: needs a key with the original singular values
- Video = **exploratory** (2 clips × 60 frames)
- Only operations that exist in the code are documented

Honesty rule: every number in this deck comes from a recorded experiment file; limits, unverified claims and corrections to the first submission are stated explicitly.

From LSB to hybrids — and an old warning about SVD schemes

1994 — LSB marking

- van Schyndel et al. [10]
- high capacity, fragile

1997 — Spread spectrum

- Cox et al. [1]
- significant spectral coefficients

2002–04 — SVD & DWT-SVD

- Liu & Tan [2]; Ganic & Eskicioglu [3]
- singular-value embedding

2010–22 — Hybrid DWT-DCT-SVD

- [7] [8] [9]
- reported gains; not re-tested here

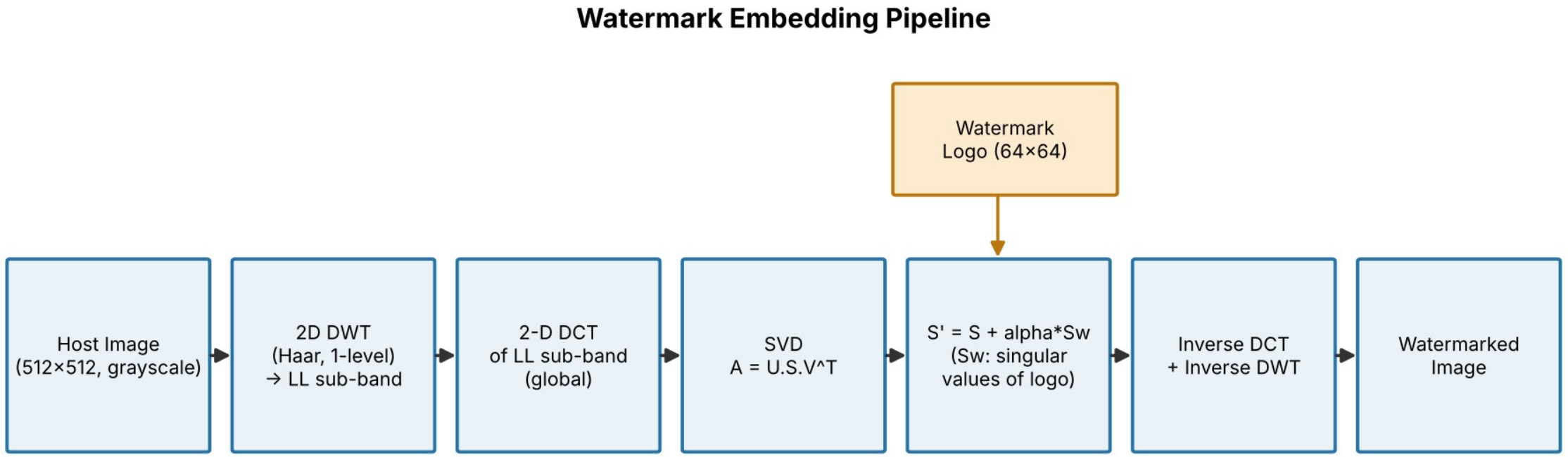
2005 — False-positive critique

- Zhang & Li [4]; Ting [5]
- key owner's mark can be 'found' anywhere

- Gap 1 — **identity binding**: SVD schemes can report the key owner's logo whatever was embedded (measured on slide 15)
- Gap 2 — **geometry**: global embedding breaks under cropping/rotation, the standard remedy being invariant or block-based domains [22]
- Gap 3 — **evaluation breadth**: published hybrids are typically shown on a few standard images; here 100

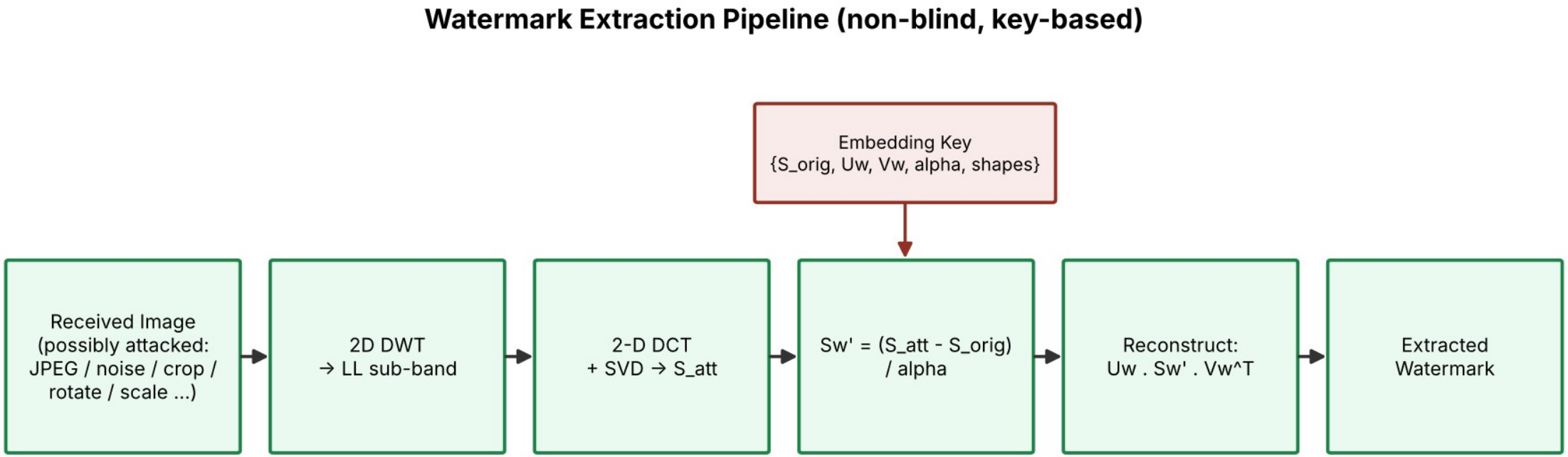
Bibliographic details verified against publisher / secondary records; status flags in research/REFERENCES.md.

Embedding: add the logo's singular values to the host's — once, by the owner



Correction vs first submission: the DCT is applied to the whole LL band (not 8×8 blocks). Alpha = 0.10 in the main experiment.

Extraction needs the owner's key — and rebuilds the owner's logo from it



Correction vs first submission: extraction is non-blind (needs S_orig of the unmarked host); the key holds U_w and V_w, i.e. the logo's identity.

One tunable parameter: add α times the logo's singular values

DWT (HAAR, LL)

$$LL = (a + b + c + d) / 2 \quad \text{for each } 2\times2 \text{ block}$$

DCT-II (GLOBAL, ORTHONORMAL)

$$F(u,v) = \alpha(u)\alpha(v) \sum\sum f(x,y) \cos[(2x+1)u\pi/2M] \cos[(2y+1)v\pi/2N]$$

SVD

$$A = U \cdot \mathrm{diag}(S) \cdot V^{T} \qquad W = U_w \cdot \mathrm{diag}(S_w) \cdot V_w^{T}$$

EMBEDDING

$$S' = S + \alpha \cdot S_w , \quad A' = U \cdot \mathrm{diag}(S') \cdot V^{T} \rightarrow \text{IDCT} \rightarrow \text{IDWT} \rightarrow \text{clip}$$

EXTRACTION (KEY: S_ORIG, U_W, V_W, A)

$$S_w' = (S_{att} - S_{orig}) / \alpha , \quad W' = U_w \cdot \mathrm{diag}(S_w') \cdot V_w^{T}$$

What follows from the maths

- **Non-blind:** needs S_orig of the unmarked host
- **Global DCT:** not tied to JPEG's 8×8 blocks — JPEG robustness comes from SVD
- **Identity in the key:** U_w, V_w rebuild *the owner's* logo from any singular values
- Metric caution: a blank white image already scores NC = **0.36** against this logo

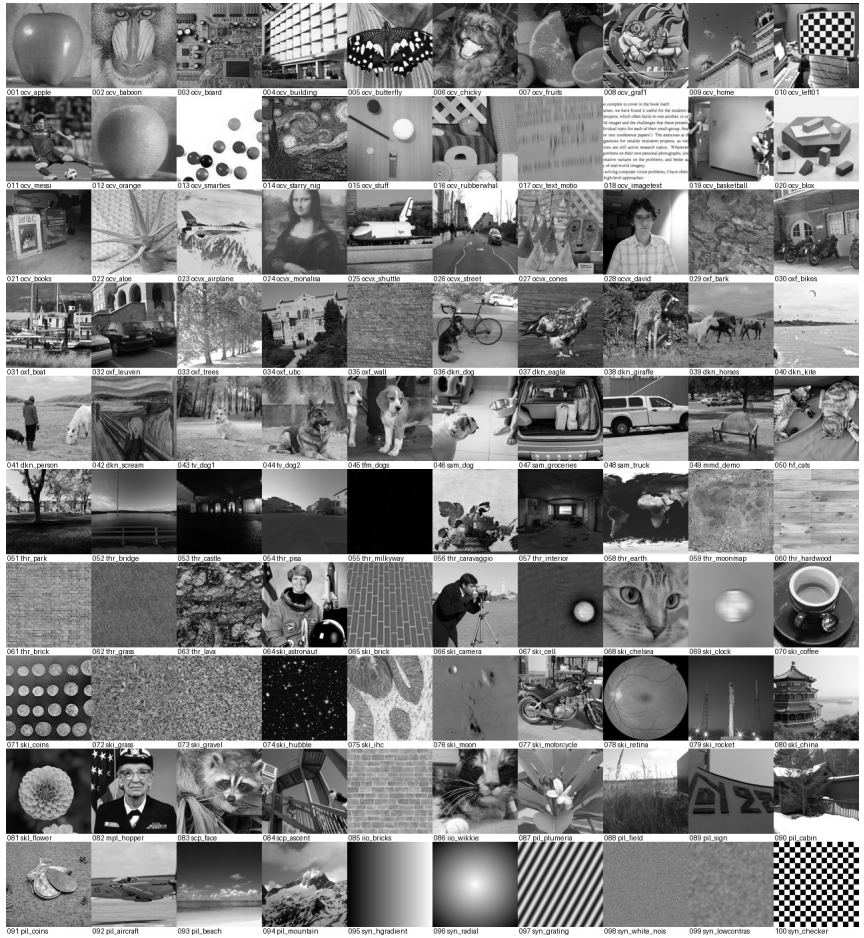
Equations (1)–(5) of the report; numerics unchanged from the first submission (re-run reproduces its table within 0.004 NC).

Twenty documented operations — only those that exist in the code

#	Transformation	Library	#	Transformation
T1	Grey conversion, centre-crop, Lanczos resize	Pillow	T11–T12	Median 3×3, Gaussian blur σ=1.5
T2	1-level Haar DWT / inverse	PyWavelets	T13–T15	Rotation 5°/15°, border cropping, 0.5× scaling
T3	2-D DCT-II / inverse (global)	SciPy	T16–T17	Histogram equalisation, unsharp mask
T4	SVD of host sub-band and logo	NumPy	T18	PSNR, SSIM, MSE
T5	Singular-value embedding and extraction	NumPy	T19	NC, ZNCC (new), BER
T6–T7	Anti-aliased resize; procedural logo	scikit-image, Pillow	T20	Frame I/O, per-frame embed, codec round trip (new)
T8	JPEG Q=90/50/20	Pillow	T9–T10	Gaussian σ^2=0.005/0.02; salt-and-pepper 2 %

Pipeline order matters: DWT → DCT → SVD → embed → **round to 8 bit** → attack → DWT → DCT → SVD → extract. The first submission attacked the float image; the 8-bit protocol is primary here (difference < 0.001 mean NC except histogram equalisation).

100 unique images: 94 real, 6 labelled synthetic — rights and provenance recorded



94
real images

6
synthetic controls

18
categories

Provenance, diversity, limits

- Sources: OpenCV (+extra), scikit-image, darknet, torchvision, three.js, Pillow tests, …
- Brightness: mid 87, dark 7, bright 6; contrast: medium 62, low 19, high 19
- SHA-256, URL, licence and rights note per image in `metadata.csv`
- Convenience sample — **not** a benchmark; colour removed; 1–4 images in several categories

Seven experiments, 14 attacks plus a reference, four metrics — all seeded and logged

ID	What	Size
E1	embed α = 0.10, save 8-bit PNG, verify	100 images
E2	15 conditions × 2 protocols	3000 trials
E3	α sweep × 4 attacks	4000 rows
E4	false-positive controls (a–d)	1600 rows
S1/S2	scheme / component benchmarks	30 imgs / 10 ops
V1	video: 2 clips × 60 frames × 6 codecs	720 frames

Metrics

- **PSNR / SSIM** vs the unwatermarked original
- **NC**, **ZNCC** (mean-centred), **BER** vs the logo
- Recognised if NC ≥ 0.75 (project convention)
- 95 % bootstrap CIs; Wilson intervals for pass rates

100/100 images · **3000/3000** attack trials · **4000/4000** sweep rows — **0 failures**

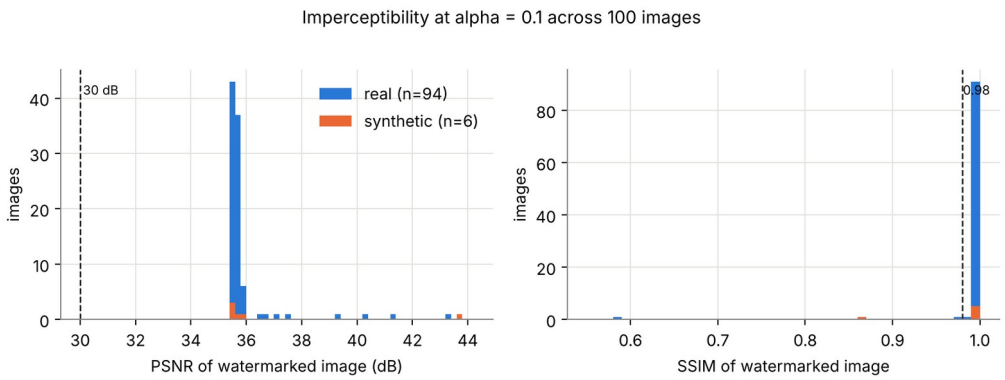
Two full runs gave **identical metrics** (max |Δ| = 0.0) · 43 tests pass · ≈ 80 s on 4 cores

Invisible on typical images — but PSNR alone hides some failures

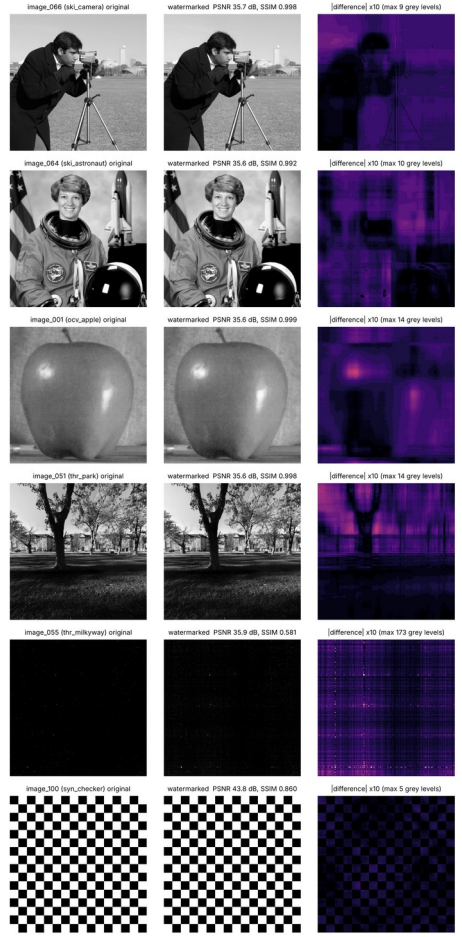
36.0 dB
mean PSNR (min 35.5)

0.992
mean SSIM · 3 images < 0.98

0
images below 30 dB

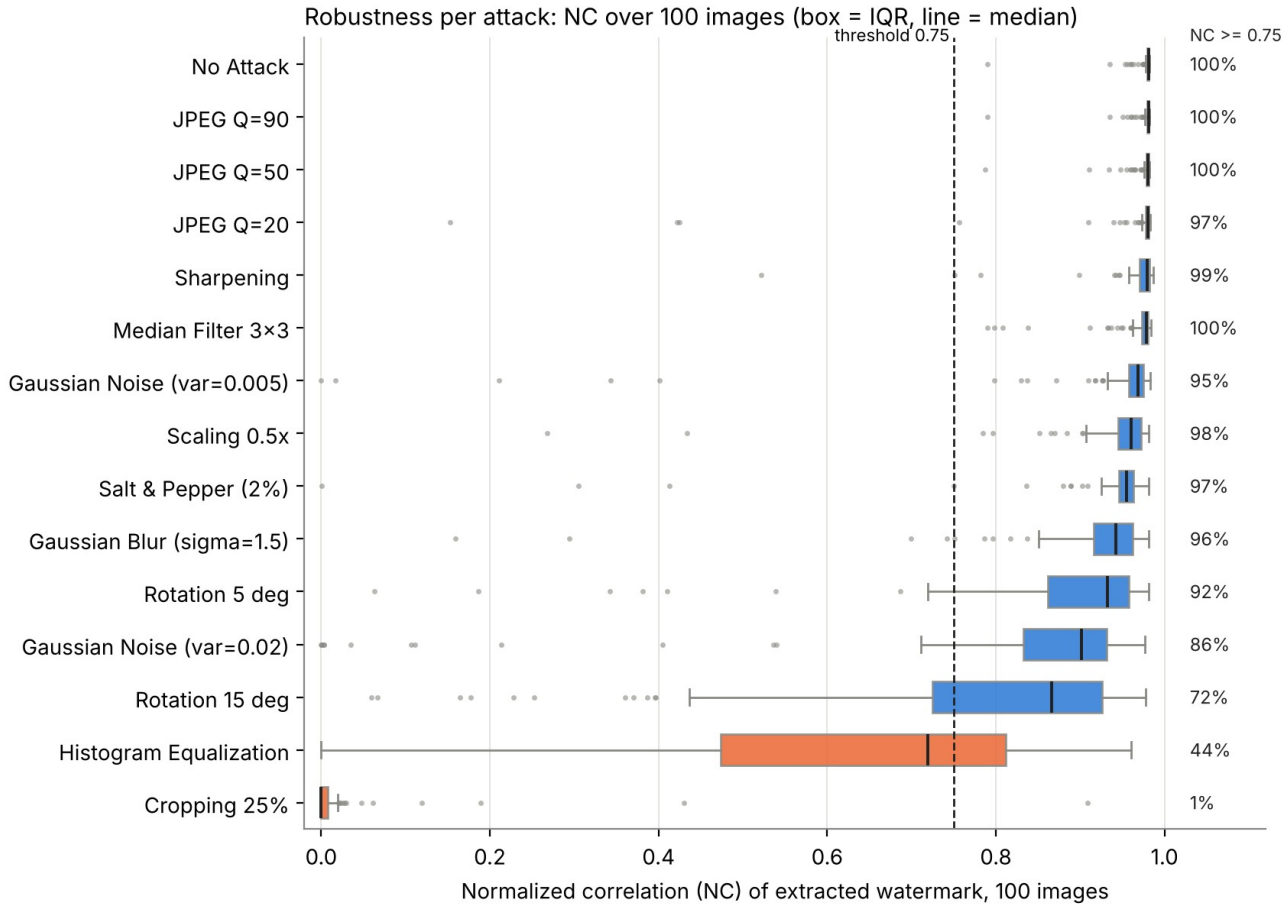


- PSNR ≈ 35.6 dB for **82 of 100** images — set by the logo, not the host
- **Near-black sky** (`image_055`): PSNR 35.9 dB but **SSIM 0.58** — clipping dominates
- Checkerboard: SSIM 0.86; synthetic images reach 43.8 dB



Right: original · watermarked · |difference|×10 for typical images and two failure cases (E1, 100 images).

85.1 % of trials recovered — strong for compression and filtering, weak for cropping and equalisation

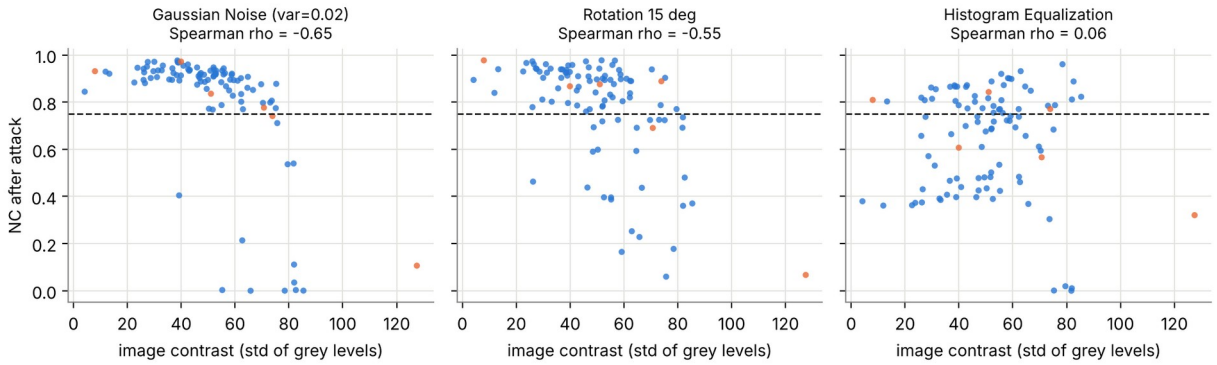


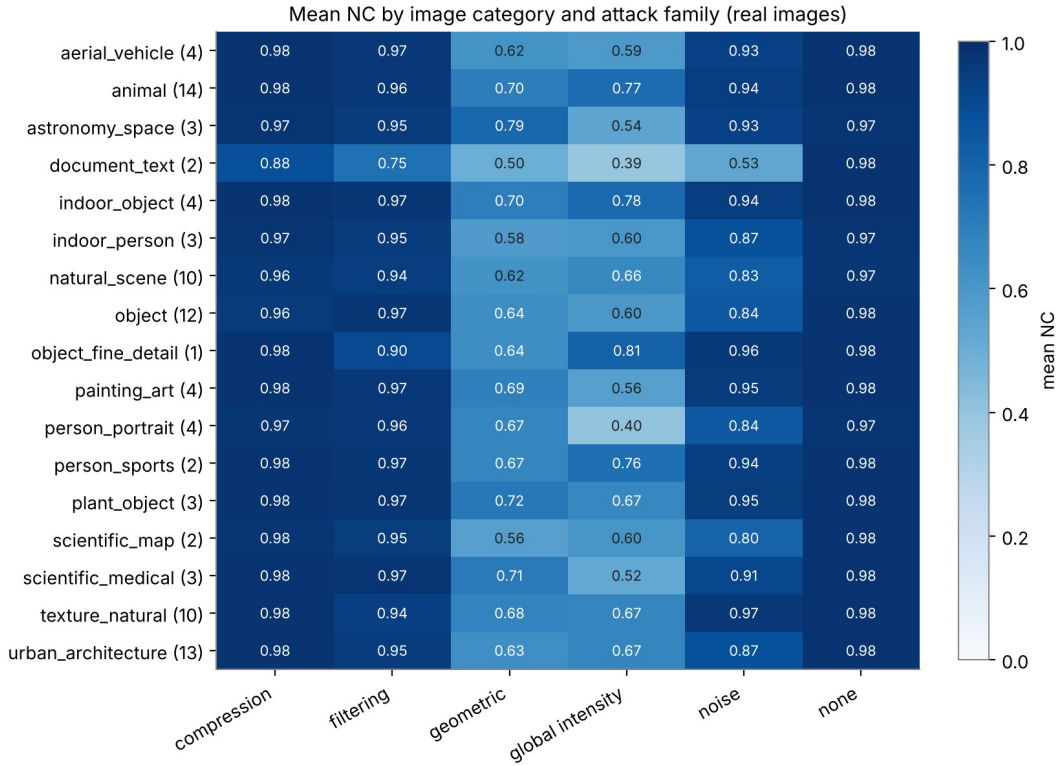
Attack	mean NC	NC ≥ 0.75
JPEG Q=50	0.975	100 %
Median 3×3	0.967	100 %
Scaling 0.5×	0.938	98 %
Noise σ^2=0.02	0.806	86 %
Rotation 15°	0.776	72 %
Hist. equalisation	0.644	44 %
Border cropping	0.022	1 %

First submission: "rotation/scaling NC > 0.93". At n = 100: scaling 0.938 ✓, **5° rotation 0.880 ×**.

1500 trials = 100 images × 15 conditions (14 attacks + reference); attacks only: 1177/1400 = 84.1 %.

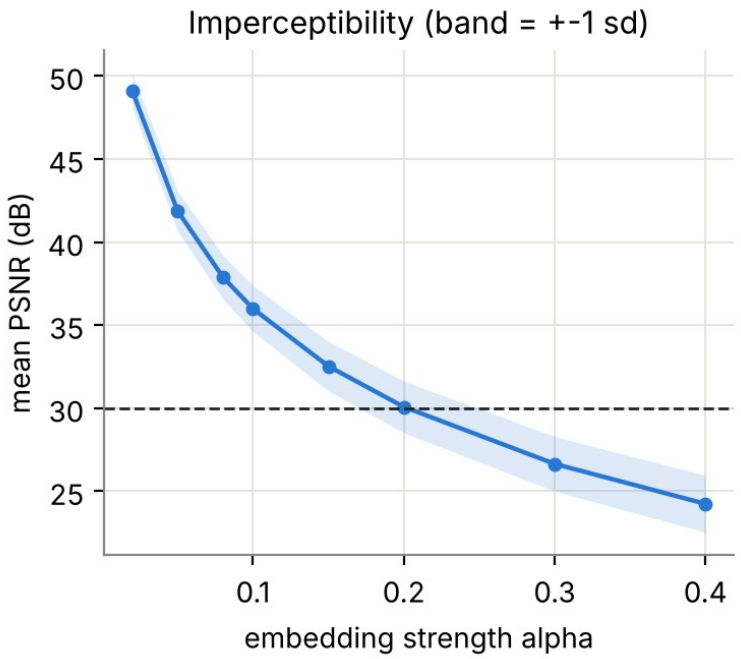
Robustness depends on the image: higher-contrast images lose more under noise and rotation

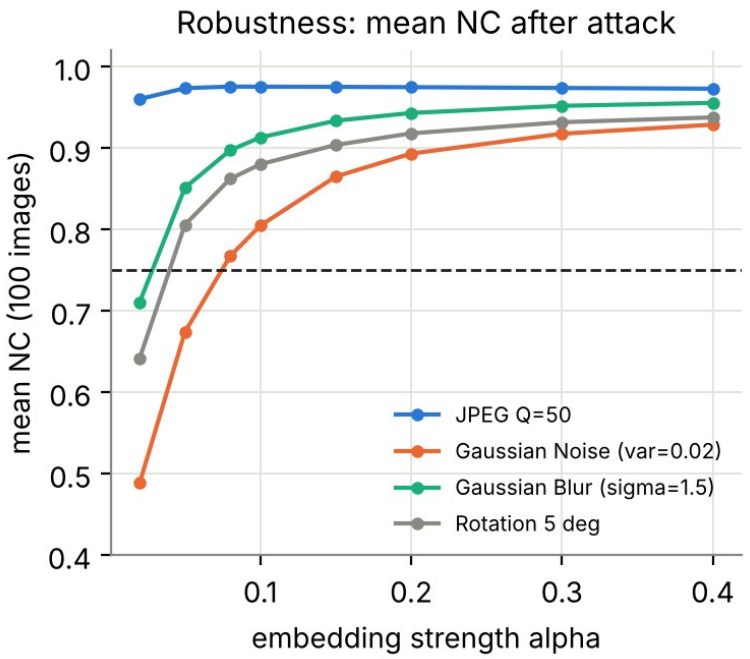


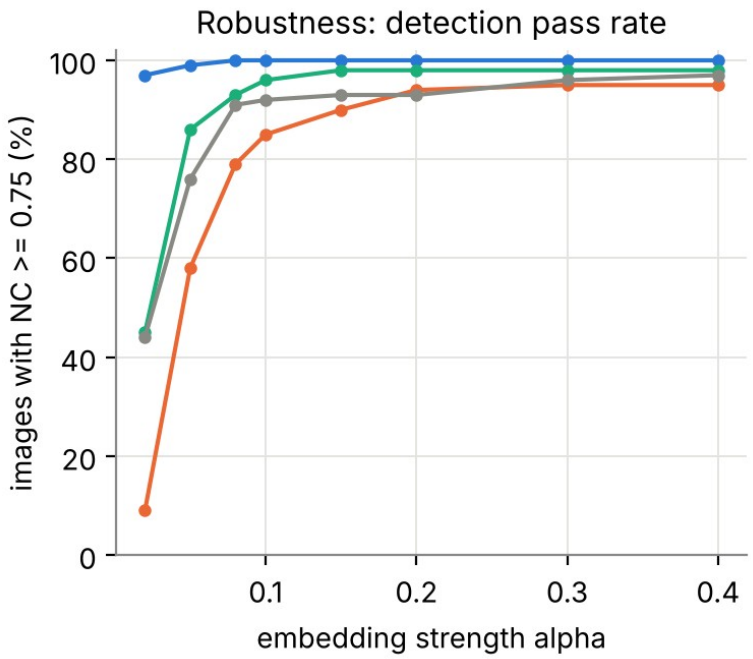


- Noise σ^2=0.02: ρ(contrast) = **-0.65**; salt-and-pepper **-0.68**; 15° rotation **-0.55**
- Hist. equalisation tracks mean brightness (ρ = -0.31), not contrast
- Weakest categories: **document text** (noise 0.53) and **portraits** under equalisation (0.40)
- n = 100, uncorrected p-values — associations, not causes

The α trade-off is real for noise and rotation — but flat for JPEG, so the earlier figure misled







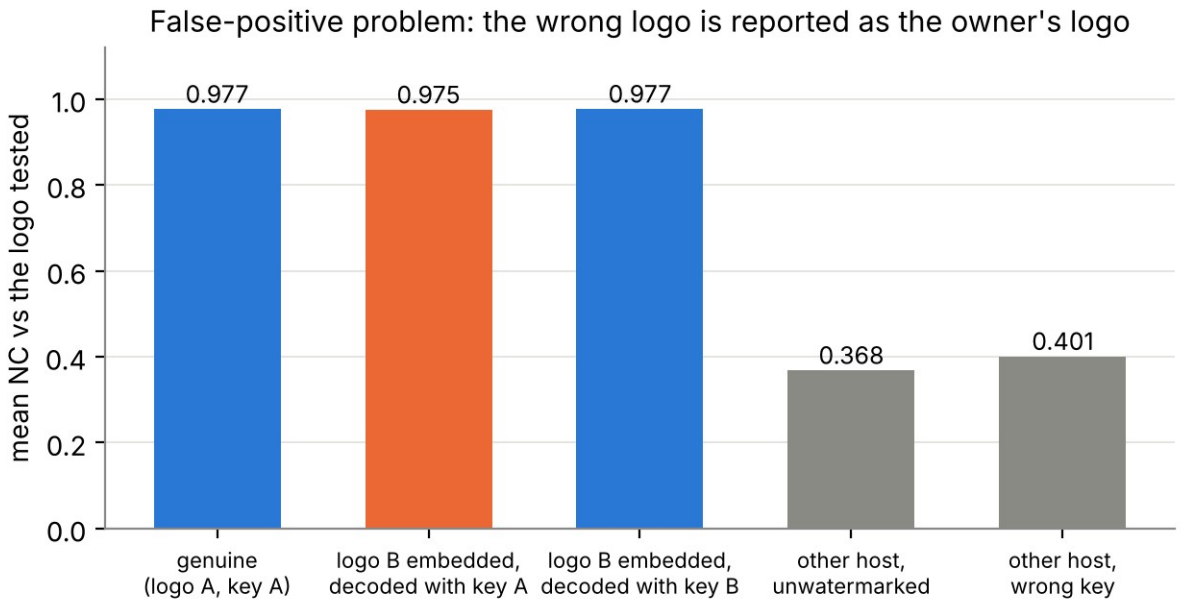
9 % → 85 %
noise pass rate, α 0.02 → 0.10

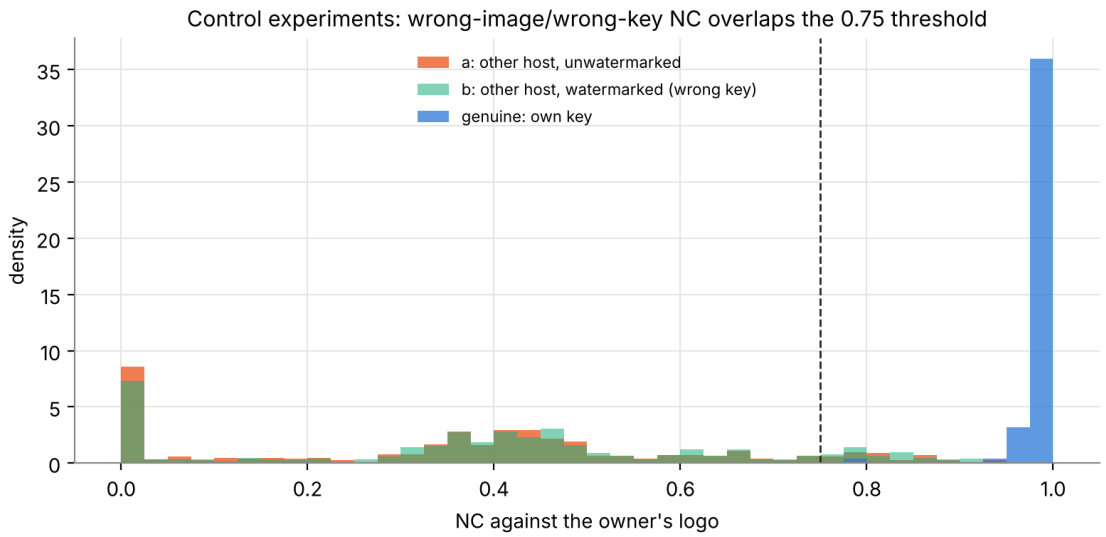
36.0 → 30.1 dB
PSNR, α 0.10 → 0.20

0.96 → 0.97
mean NC after JPEG Q=50, α 0.02 → 0.40

Old figure: axis 0.93–0.99 + float truncation before JPEG → **artefact** (`exp_jpeg_truncation.py`)

A different logo decoded with the owner's key still scores NC 0.98 — NC does not prove ownership





0.975
wrong logo, owner's key (genuine 0.977)

9.8 % / 12.6 %
unrelated-image controls above NC 0.75

0.82
95th percentile NC of unrelated images

Needed: an **image-bound signature** — not implemented; the principal-component variant was **worse** (NC 0.98 on an unmarked image)

Four failure modes — explained from the data, not hidden

Border cropping

- mean NC **0.022**, pass **1 %**
- 43.75 % of the *area* zeroed (25 % per dimension)
- all singular values change; logo extracts as black (BER 0.14 ≈ white fraction)

Histogram equalisation

- mean NC **0.644**, pass **44 %**
- remaps the whole intensity distribution
- varies by image (first report: 0.10 vs 0.66)
- tracks mean brightness (ρ −0.31)

Rotation > 5°

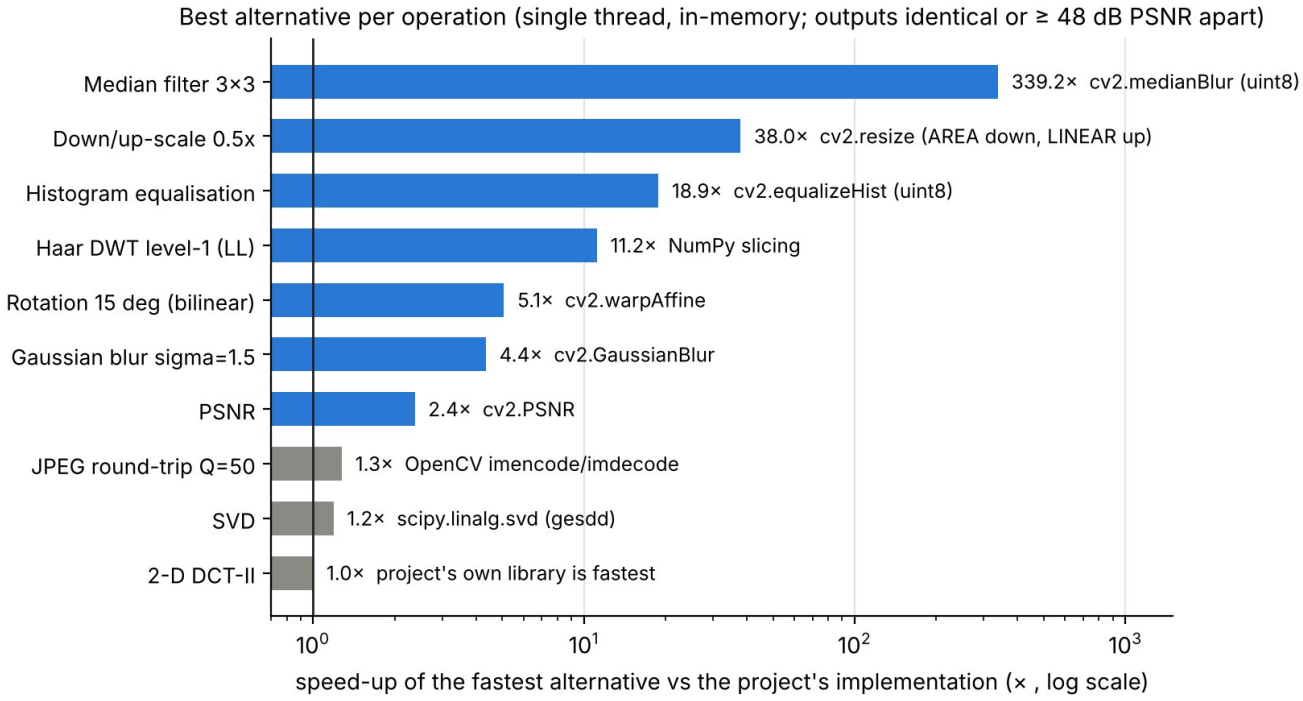
- 5°: **92 %**, 15°: **72 %**
- no re-registration attempted
- consistent with RST theory [22]
- block-based schemes break too (blind-watermark BER 0.21)

Near-constant images

- near-black sky: PSNR **35.9 dB**, SSIM **0.58**
- checkerboard: lowest clean NC (0.79)
- PSNR alone would have hidden both

By contrast: JPEG Q ≥ 20, median 3×3, sharpening, 0.5× scaling, blur, salt-and-pepper and mild noise all reach ≥ 95 % pass rate.

Same operation, different libraries: numerically equal, speed rarely matters here

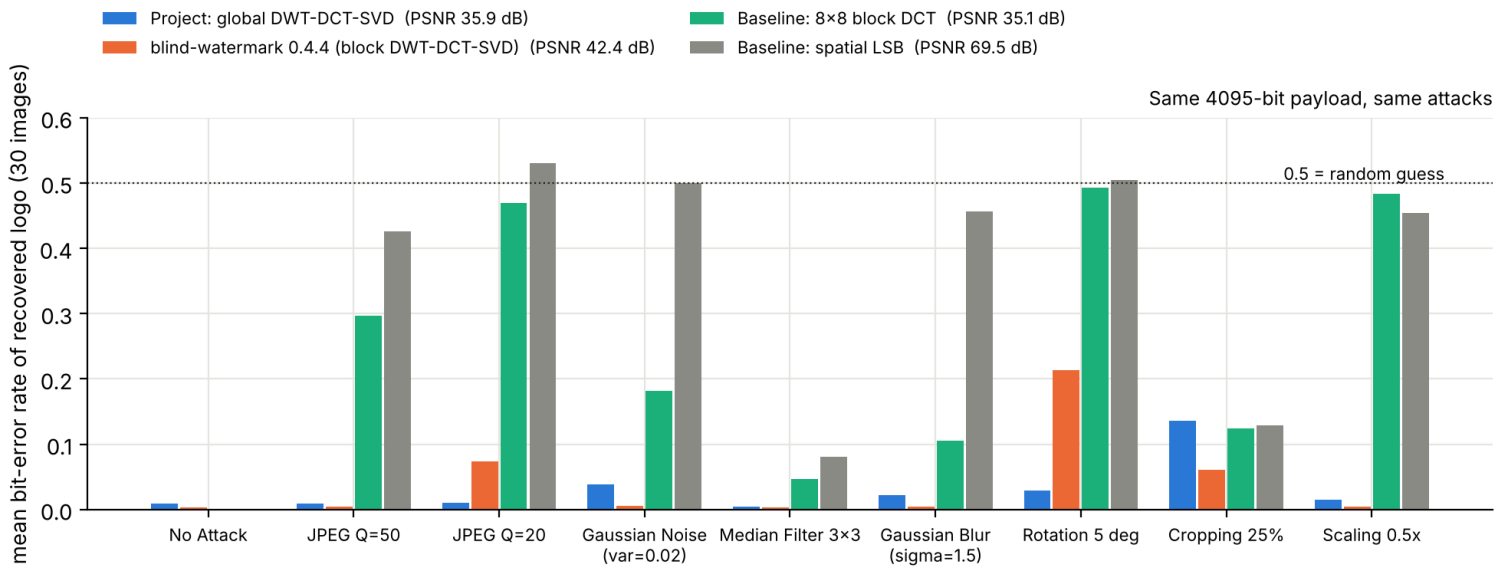


What the measurements say

- DCT / SVD / Haar agree to ≤ 1.5e-16 (DWT 1e-13); Pillow ≡ OpenCV JPEG
- SVD dominates embedding (≈ 23 of 37 ms); `scipy.linalg.svd` is **1.19×** NumPy's
- OpenCV: median **≈ 340×**, blur 4.4×, rotation 5.1×, rescale 38×, equalise 19× (partly `uint8` vs `float64`)
- PyWavelets Haar is **11× slower** than four lines of NumPy — still the right general tool
- ImageMagick CLI ≈ **49 ms** per JPEG (process + disk)
- Whole experiment ≈ **80 s** on 4 cores → no performance case for switching

Documentation-based capability matrix (not benchmarked) is in the report, Table 3.1.

Beats LSB and block-DCT at equal PSNR — but not an existing blind block-based hybrid



	Project	blind-watermark	block DCT	LSB
PSNR (dB)	35.9	42.4	35.1	69.5
JPEG Q=50 BER	0.010	0.004	0.296	0.426
Rotation 5° BER	0.029	0.214	0.493	0.504
Cropping BER	0.135	0.060	0.124	0.128
embed / extract ms	32 / 17	420 / 315	26 / 12	2 / 1
blind?	**no** (key)	yes	yes	yes

Reading the comparison

- Project ≫ LSB and block-DCT under JPEG, scaling, rotation
- `blind-watermark`: lower BER under JPEG Q=50 (0.004 vs 0.010), noise, blur, cropping; **higher PSNR** (42.4 dB)
- …but **13× slower** and worse under rotation (0.214 vs 0.029) and JPEG Q=20
- Its colour/YUV design and channel-wise attacks may favour it
- **No claim of superiority** — contribution = explainable reference implementation + measured weaknesses

Per-frame embedding survives H.264 and MJPEG re-encoding — feasibility, not a video benchmark

Codec setting	PSNR (dB)	mean NC	min NC	size (kB)
lossless (FFV1)	35.6	0.980	0.977	5289
H.264 CRF 18	35.0	0.980	0.977	381
H.264 CRF 23	34.6	0.980	0.976	181
H.264 CRF 28	33.9	0.979	0.974	88
MJPEG q=5	34.3	0.979	0.975	1244
MJPEG q=15	32.3	0.977	0.972	595

Embedding ≈ 32–41 ms per 512^2 frame on one core (≈ 24–31 frames/s) · clips: `vtest.avi` (768×576), `Megamind.avi` (720×528) from the OpenCV sample data

720
decoded frames, 0 failed

≥ 0.97
NC in every frame

Caveats

- 2 clips × 60 frames — small
- one **key per frame** (≈ 2 MB)
- no frame dropping, re-timing, colour or hardware codecs
- naive frame-drop still NC 0.97 (similar neighbours)

What held up and what did not: ten claims of the first submission checked at n = 100

First submission	This evaluation (100 images)	Verdict
PSNR 35.6 dB / SSIM 0.996 (2 images)	36.0 dB / 0.992; 3 images < 0.98	✓ with exceptions
25 / 30 trials ≥ 0.75	1277 / 1500 (85.1 %); attacks only 84.1 %	✓
JPEG: NC > 0.97	0.977 / 0.975 / 0.956 (Q 90/50/20)	✓ (Q ≥ 50)
Rotation/scaling NC > 0.93	scaling 0.938; rotation 5° 0.880	scaling ✓ · rotation ×
Cropping fails; hist-eq 0.10 / 0.66	pass 1 %; hist-eq mean 0.644 (44 % pass)	✓ (hist-eq varies)
Robustness rises with α (JPEG)	flat 0.960 → 0.973	× artefact
"Block DCT", "semi-blind"	global DCT, non-blind	corrected
"Fifteen attacks"	14 attacks + the unattacked reference	corrected
Extraction proves ownership	wrong logo decoded at NC 0.975	× **not supported**
Hybrids beat single transforms	not tested; blind-watermark beats the project	unverified

Tested, seeded and logged: 43 tests, zero failures, identical re-runs

Engineering

- numerics of `embed/extract_watermark` unchanged
- validation: 2-D, finite, size, α > 0
- `image_io`: alpha, 16-bit, corrupt/empty files
- config-driven batch runner, 4 workers

Verification

- **43 tests** pass (functions, edge cases, loaders, completeness)
- legacy re-run reproduces the first report (≤ 0.004 NC)
- two full runs: identical metrics
- all 100 output PNGs re-opened and validated

Traceability

- originals preserved in `original_backup/`
- CHANGELOG lists every change and why
- negative results saved: principal-component variant; JPEG truncation
- PROJECT_AUDIT: 15 findings, F1–F15

REPRODUCE

```
python build_dataset.py → run_batch_experiment.py → analysis.py → benchmark_tech.py → video_demo.py
python -m pytest tests -q · Python 3.13, NumPy 2.5, SciPy 1.18, scikit-image 0.26, Pillow 12.3, OpenCV 5.0, FFmpeg 6.1
```

Known limits — and the next steps the evidence supports

Limitations (measured)

- **False-positive problem** — not fixed
- **Non-blind**: ≈ 2.1 MB key per image
- **Cropping** 1 %, **rotation 15°** 72 %, **hist-eq** 44 %
- Grey-scale only; clean BER ≈ 0.9 % (logo resampling)
- Dataset: convenience sample, rights mostly undocumented
- Only one existing package compared; no GPU; ±5–10 % timing noise
- Video: exploratory, per-frame keys, no temporal attacks

Future scope (in priority order)

- **1 · Image-bound signature** to remove false positives — verify with the E4 controls
- **2 · Block-based redundant embedding** for cropping; use `blind-watermark` as the new baseline
- **3 · Geometric re-synchronisation** (log-polar / template) [22]
- **4 · Colour:** embed in the Y channel of YCbCr
- **5 · Video:** index-aware keys, temporal redundancy, frame-drop and re-timing tests; real-time port (≈ 27 fps single-core as is)
- **6 ·** Learned watermarkers [29][30] as a comparison, if the no-ML rule is relaxed

A transparent scheme that survives everyday processing — but NC alone is not proof of ownership

36.0 dB / 0.992

mean PSNR / SSIM over 100 images

85.1 %

of 1500 trials with NC ≥ 0.75

NC 0.98

for a DIFFERENT logo with the owner's key

KEY TAKEAWAYS

- **Works:** JPEG Q ≥ 20, median, sharpening, scaling, blur, mild noise — ≥ 95 % pass rate; per-frame video re-encoding too
- **Does not:** border cropping (1 %), histogram equalisation (44 %), rotation 15° (72 %); first report's α-trade-off figure was an artefact for JPEG
- **Cannot claim:** ownership from NC alone (false-positive problem) or superiority over existing technology
- **Next:** bind the logo to the image, block-based redundancy, geometric re-synchronisation

Verified sources (IEEE style) — status flags in research/REFERENCES.md

[1] I. J. Cox et al., “Secure spread spectrum watermarking for multimedia,” IEEE Trans. Image Process., 6(12):1673–1687, 1997.

[2] R. Liu and T. Tan, “An SVD-based watermarking scheme for protecting rightful ownership,” IEEE Trans. Multimedia, 4:121–128, 2002.

[3] E. Ganic and A. M. Eskicioglu, “Robust DWT-SVD domain image watermarking: embedding data in all frequencies,” Proc. MM&Sec ’04, ACM, pp. 166–174, 2004.

[4] X.-P. Zhang and K. Li, “Comments on ‘An SVD-based watermarking scheme…’,” IEEE Trans. Multimedia, 7:593–594, 2005.

[5] G. C.-W. Ting, “Ambiguity attacks on the Ganic–Eskicioglu DWT-SVD scheme,” Proc. ICISC, pp. 378–388, 2005.

[7] C.-C. Lai and C.-C. Tsai, “Digital image watermarking using DWT and SVD,” IEEE Trans. Instrum. Meas., 59(11):3060–3063, 2010.

[8] S. Bekkouch and K. M. Faraoun, “…combined DCT-DWT-SVD transforms,” J. Inf. Process. Syst., 11(3):406–420, 2015.

[9] M. Begum, J. Ferdush, M. S. Uddin, “A hybrid robust watermarking system based on DCT, DWT and SVD,” J. King Saud Univ. CIS, 34(8):5856–5867, 2022.

[10] R. G. van Schyndel et al., “A digital watermark,” Proc. ICIP, vol. 2, pp. 86–90, 1994. [11] E. Koch and J. Zhao, Proc. IEEE NSIP Workshop, pp. 452–455, 1995.

[12] S. G. Mallat, IEEE Trans. PAMI, 11(7):674–693, 1989. [13] N. Ahmed, T. Natarajan, K. R. Rao, IEEE Trans. Comput., C-23(1):90–93, 1974.

[16] Z. Wang et al., “Image quality assessment: from error visibility to structural similarity,” IEEE Trans. Image Process., 13(4):600–612, 2004.

[17] F. Hartung and M. Kutter, “Multimedia watermarking techniques,” Proc. IEEE, 87(7):1079–1107, 1999.

[22] C.-Y. Lin et al., “Rotation, scale, and translation resilient watermarking for images,” IEEE Trans. Image Process., 10(5):767–782, 2001.

[23] F. Hartung and B. Girod, “Watermarking of uncompressed and compressed video,” Signal Process., 66(3):283–301, 1998.

[28] `blind-watermark` 0.4.4, github.com/guofei9987/blind_watermark · [29] J. Zhu et al., HiDDeN, ECCV 2018 · [30] M. Tancik et al., StegaStamp, CVPR 2020

Software: NumPy [19], SciPy [20], scikit-image [21], PyWavelets [18], Pillow, OpenCV, FFmpeg, ImageMagick. Dataset sources and attributions: data/README.md.